# GridSystem Class

**Namespace:** `DT.GridSystem`
**Inherits from:** `UnityEngine.MonoBehaviour`

A generic 2D grid container that stores objects in a 1D array and provides utilities for grid manipulation, coordinate conversion, and world-to-grid mapping.

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## Description

`GridSystem<TGridObject>` manages a 2D grid used for gameplay or data mapping.
Objects are stored in a **row-major array**, and accessed using `(x,y)` coordinates, world positions, or flat indices.
Derived classes implement world-coordinate mapping behavior.

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## Properties

| Property | Type | Description |
|---|---|---|
| `GridSize` | `Vector2Int` | Current grid dimensions `(width, height)` in cells. |
| `CellSize` | `float` | World size of each grid cell. |
| `OnGridUpdated` | `UnityEvent` | Invoked whenever grid data or structure changes. |

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## Public Methods

### Initialization & Setup

| Signature | Description |
|---|---|
| `void SetUpGrid(Vector2Int size, float cellSize)` | Initializes or resizes grid. Reallocates storage if needed. |

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## Object Placement

| Signature | Description |
|---|---|
| `void AddGridObject(int x, int y, TGridObject value, bool snapToGrid = false)` | Adds or replaces object at grid coordinate. |
| `bool AddGridObject(Vector3 worldPos, TGridObject value, bool snapToGrid = false)` | Adds object based on world position; returns false if outside grid. |

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## Object Removal

| Signature | Description |
|---|---|
| `TGridObject RemoveGridObject(int x, int y)` | Removes and returns object at cell; returns default if invalid. |
| `bool RemoveGridObject(Vector2Int pos, out TGridObject removed)` | Removes and outputs object; returns true on success. |

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## Object Retrieval

| Signature | Description |
|---|---|
| `TGridObject GetGridObject(int x, int y)` | Returns object at cell; returns default if invalid. |
| `bool TryGetGridObject(int x, int y, out TGridObject obj)` | Attempts to fetch object; returns true if valid. |
| `bool TryGetGridObject(Vector3 worldPos, out TGridObject obj)` | Attempts lookup from world position; returns true if valid. |
| `TGridObject GetOrNull(int x, int y)` | Safe getter that returns default when out of bounds. |

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## Coordinate Conversion

| Signature | Description |
|---|---|
| `int ToIndex(int x, int y)` | Converts `(x,y)` to flat array index. |
| `void FromIndex(int index, out int x, out int y)` | Converts flat index into `(x,y)` coordinates. |
| `int WorldToIndex(Vector3 worldPosition)` | Returns index for world position; returns -1 if outside grid. |

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## World ↔ Grid Mapping

> *(implemented in derived classes)*

| Signature | Description |
|---|---|
| `abstract Vector3 GetWorldPosition(int x, int y, bool snapToGrid = false)` | Converts grid coordinate to world position. |
| `abstract void GetGridPosition(Vector3 worldPosition, out int x, out int y)` | Converts world position to grid coordinate. |
| `Vector3 SnapWorldPosition(Vector3 worldPosition)` | Snaps world position to center of nearest grid cell. |
| `bool IsInBounds(Vector2Int pos)` | Returns true if coordinate is within grid. |
| `bool IsInBounds(Vector3 worldPosition)` | Returns true if world position lies in bounds. |

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## Utility & Iteration

| Signature | Description |
|---|---|
| `void ClearGrid()` | Clears all objects (keeps dimensions). |
| `void ForEachCell(Action<int,int,TGridObject> action)` | Calls callback for every cell. |
| `IEnumerable<(Vector2Int pos, TGridObject obj)> Each()` | Enumerator of all positions and objects. |
| `Vector3 IndexToWorld(int index)` | Returns world position of cell at index. |

## Editor-Only Methods *(Unity Editor only)*

| Signature | Description |
|---|---|
| `void ResizeGridIfNeededEditor()` | Resizes internal array when grid size changes in Inspector. Includes Undo and value preservation. |
| `void OnDrawGizmos()` | Draws visual grid in Scene View. |